

COMPLETE 1:1 TRUE-SCALE PCB FABRICATION SHEET • COPPER ETCH MASKS & DRILL TEMPLATE (REV 2.1)

[8.Cu SOLDER SIDE: SOLID BLACK COPPER TRACES & PADS FOR ETCHING]

[F.Cu COMPONENT SIDE: SIGNAL & POWER TRACES]

100.0 mm TRUE SCALE CALIBRATION RULER (MEASURE WITH CALIPER BEFORE ETCHING)

DRILL BIT APERTURES & HOLE SIZES:

- 3.20mm (M3 Drill): 5x Holes (Chassis Mounting Standoffs)
- 1.40mm (0.055 in): 7x Holes (16A 5.08mm Screw Terminals TB1, TB2, TB3)
- 1.00mm (0.040 in): 52x Holes (Arduino Mega 2560 Header Sockets)
- 0.99mm (0.039 in): 32x Holes (JST-XH 2.50mm Shrouded Ultrasonic Headers)
- 0.85mm (0.033 in): 6x Holes (1000uF Low-ESR Radial Capacitors C1, C2, C3)
- 0.80mm (0.031 in): 22x Holes (PCB17 Opto U1, Servos J_SRV1-6, HX711)
- 0.40mm (0.016 in): 18x Vias (Inter-layer Copper Stitching Vias)

CHEMICAL ETCHING & TONER TRANSFER SOP:

1. Print Panel 1 on glossy magazine paper / heat-transfer film at 100% scale (no fit).
2. Clean double-sided 1.6mm FR-4 copper clad with isopropyl alcohol & scouring pad.
3. Iron toner onto copper at 200°C for 4.5 mins. Soak in water & gently peel paper.
4. Etch in ferric chloride (FeCl3) solution at 45°C with agitation until clear.
5. Rinse thoroughly, strip toner with acetone, apply liquid tin or flux coating.